

Comment of the Obsidian Watch Group

Docket: HHS-OASH-2026-0232

Request for Information: Temporary Placement of 7-Hydroxymitragynine Above a

Specified Threshold in Schedule I Comment period closes: 10 September 2026

Submitted by: Jay Puckett, Principal Security Researcher, Obsidian Watch Group

SUMMARY OF POSITION

We do not oppose the temporary scheduling of 7-hydroxymitragynine (7-OH). Having reviewed the supporting record, we think the pharmacological case is sound and the 0.05% threshold has a real basis in published measurement of the plant.

We are commenting because three specific, checkable problems in the supporting record should be resolved before the threshold is finalized, and because the action as framed leaves the larger share of the documented consumer harm completely unaddressed.

We are an independent investigative group. We have no financial interest in kratom, 7-OH, or any company named in this comment, and we have received no funding from any party to this proceeding.

1. THE CITATION FOR THE THRESHOLD FIGURE DOES NOT RESOLVE

FDA's assessment, "7-Hydroxymitragynine (7-OH): An Assessment of the Scientific Data and Toxicological Concerns Around an Emerging Opioid Threat," supports the key sentence as follows:

| *"In another analysis of 13 commercial products purported to contain kratom, the*

7-OH content by weight ranged from 0.01-0.04% (Kikura-Hanajiri et al., 2009) a finding in agreement with others that have reported 7-OH to account for less than 0.05% by weight, substantially lower than reported mitragynine amounts (Kruegel et al., 2019)."

The phrase "less than 0.05% by weight" is the figure that becomes the operative legal threshold. It is attributed to Kruegel et al., 2019.

There is no Kruegel 2019 entry in the assessment's reference list. The bibliography contains Kruegel et al. 2016 (J Am Chem Soc 138(21), 6754-6764) and lists A.C. Kruegel as a co-author

on one further 2016 entry. In-text, the assessment cites "Kruegel et al., 2016" twice and "Kruegel et al., 2019" once, and the 2019 instance is the one carrying the threshold number.

We think this is most likely a citation error rather than an absent basis, because Kikura-Hanajiri et al. 2009 independently reports 0.01-0.04% and Sharma et al. 2025 reports a mean of 0.01% across 341 consumer samples. But a rule that turns on a specific number should not rest, on its face, on a reference that cannot be located.

REQUESTED ACTION: Identify the intended source for "less than 0.05% by weight" and correct the assessment, or state on the record which citations the threshold actually rests on.

2. THE THRESHOLD DOES NOT CLEANLY SEPARATE LEAF FROM CONCENTRATE

The assessment reports, from Sharma et al. 2025, that across 341 consumer-supplied kratom samples the 7-OH content ranged from below the limit of quantitation (<0.005%) to a maximum of 0.21%, with a mean of 0.01%.

A maximum of 0.21% is more than four times the proposed threshold, in ordinary consumer product. The threshold is therefore not a perfect proxy for

| *"manufactured concentrate."*

REQUESTED ACTION: State what proportion of naturally derived leaf product is expected to exceed 0.05%, and what enforcement posture applies to a botanical product that exceeds the threshold without any synthetic enrichment. Testing methodology, sampling basis and measurement uncertainty should be specified, since at these concentrations the analytical margin is a substantial fraction of the limit itself.

3. THE SUPPORTING RECORD INCLUDES INDUSTRY-FUNDED RESEARCH THAT IS NOT FLAGGED

The assessment cites Huestis, M.A., Brett, M.A., Bothmer, J., and Atallah, R. (2024), "Human Mitragynine and 7-Hydroxymitragynine Pharmacokinetics after Single and Multiple Daily Doses of Oral Encapsulated Dried Kratom Leaf Powder," *Molecules* 29(5).

That paper's published disclosure states that Johnson Foods LLC paid for the clinical study on its own dried kratom leaf powder product, and that all four authors are paid consultants to Johnson Foods, LLC.

Johnson Foods, LLC is a registered federal lobbying client on kratom issues, with approximately \$130,000 in disclosed lobbying reported through Holland & Knight in Senate Lobbying Disclosure Act filings.

One co-author, R. Atallah, is publicly identified as Chief Science and Research and Development Officer at Botanic Tonics, LLC, a manufacturer of leaf-kratom tonic products, and as a consultant to the American Kratom Association. We note that the Botanic Tonics role may postdate the February 2024 publication, and we make no allegation that any disclosure obligation was breached.

We raise this only because a rule that restricts one product category while leaving a competing category unrestricted should be able to show that its evidence base was assessed for interest.

REQUESTED ACTION: State whether the funding and consulting relationships in the cited literature were evaluated, and whether any cited finding is load-bearing for the threshold determination.

4. THE ACTION LEAVES THE LARGER DOCUMENTED HARM UNADDRESSED

FDA's own CFSAN Adverse Event Reporting System contains 879 reports naming kratom products. We recognize that CAERS reports are voluntary, unverified, and do not establish causation. With that caveat, the reported outcomes are:

Death	276
Hospitalization	239
Disability	135
Life threatening	134

Other serious or important medical event 427

Emergency room visit	95
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The most frequently reported reaction after death is dependence, at 256 reports, followed by withdrawal syndrome at 98. Reports rose to 151 in 2025. The largest affected age group is 25 to 34, with 278 reports, and 11 reports involve a person under 18.

The great majority of these reports concern leaf and mitragynine products, which this action does not reach. Sixty-two reports name a single leaf-kratom tonic brand and include four deaths, nineteen hospitalizations, sixteen disabilities and eleven life-threatening events, with dependence the most frequently reported effect at thirty-six.

We understand that mitragynine has been granted Investigational New Drug status and that NIDA is sponsoring a Phase 1 trial of a purified mitragynine formulation for opioid use disorder. We think that is a legitimate reason not to place mitragynine in Schedule I, and we are not asking the Department to do so.

We are asking the Department not to allow the 7-OH action to stand as the whole of the federal consumer-protection response.

REQUESTED ACTION: State what consumer protection measures apply to mitragynine-containing retail products that this action does not reach, including labeling of alkaloid content, dosing information, and disclosure of dependence and withdrawal risk.

5. FIRSTHAND ACCOUNT: DEPENDENCE ONSET AND MEDICALLY MANAGED WITHDRAWAL

The Department's own assessment states that "a well-designed human abuse potential study has not been conducted that would show whether kratom, mitragynine, or 7-OH produce rewarding effects." This Request for Information is therefore the appropriate place to submit direct user experience, and we offer one account.

The undersigned took a commercially available 7-OH product deliberately, in order to test the claim that these products produce dependence. He had no prior opioid dependence.

The product was Limitless Black Edition 7-hydroxymitragynine chewable tablets, labeled at 50 mg of 7-OH per tablet, 500 mg per 10-count bottle, with a labeled serving size of one quarter tablet, that is 12.5 mg.

- Total consumed: 6,000 mg of 7-OH. That is 120 tablets, twelve bottles, over approximately three weeks: an average of roughly 286 mg per day, or about 23 labeled servings per day. - Physical dependence developed within the first seven days. Use continued for approximately three weeks in total.

- Withdrawal was medically managed with prescribed buprenorphine, a partial mu-opioid agonist indicated for opioid withdrawal. A licensed prescriber assessed the presentation and treated it as opioid withdrawal. - The dose was 8 mg tablets, 3.75 tablets in total, a cumulative 30 mg of buprenorphine across the withdrawal course. - Contemporaneous toxicology was positive for mitragynine and for THC, and negative for all other opioids.

Prescription and laboratory documentation is retained and is available to the Department for verification on request.

WHY WE SUBMIT THIS

The routine response to kratom-related harm data is that it reflects polysubstance use. That response has substantial empirical support: reporting based on Florida medical examiner records found other substances present in approximately 96 percent of kratom-involved decedents. It does not apply here. This account involves no other opioid, and the withdrawal was severe enough to require a prescription medication indicated for opioid withdrawal.

This is directly responsive to the Department's stated evidentiary gap. It speaks to onset interval and to dependence liability in a person with no prior opioid exposure, which is the population the existing record says least about.

LIMITATIONS, STATED PLAINLY

1. This is a single case. It cannot establish a population rate and we do not offer it as one.
2. Toxicology cannot distinguish a 7-OH product from leaf kratom, because 7-OH is both a minor plant constituent and a human metabolite of mitragynine. The product identity rests on what was taken, not on the assay.
3. THC was present and is disclosed here for completeness. THC does not produce opioid withdrawal and is not treated with buprenorphine, and we do not contend it is a confound for the finding above.
4. The subject is the author of this comment and is not a disinterested party.
5. Escalation from the labeled serving to roughly 23 servings per day occurred within three weeks. We present that as the reported course, not as a claim about what any other user would do.
6. The cumulative buprenorphine dose was modest. Routine opioid use disorder maintenance commonly runs 8 to 24 mg per day; 30 mg across an entire withdrawal course is consistent with a short taper following limited-duration dependence rather than with an established opioid habit.

SEPARATELY, AND INDEPENDENT OF THIS ACCOUNT

FDA's CFSAN Adverse Event Reporting System contains four reports naming Limitless 7-hydroxymitragynine products, two of them Black Edition specifically. Recorded outcomes include three "Other Serious or Important Medical Event" and one Disability. Reported reactions include withdrawal syndrome, restless legs syndrome, hyperhidrosis, insomnia, anxiety and dehydration, a symptom cluster consistent with opioid withdrawal. Ages reported were 23, 37, 56 and 69.

The seven warning letters FDA issued on 15 July 2025 went to Shaman Botanicals, My Smoke Wholesale, Relax Relief Rejuvenate Trading, Thang Botanicals, Royal Diamond Imports, Hydroxie, and 7Tabz Retail. Limitless is not among them. A product labeled at 50 mg of 7-OH per tablet, carrying a disability outcome in the Department's own reporting system, has received no warning letter and remains on the market.

REQUESTED ACTION: The Department has stated that no human abuse potential study for these substances exists. We ask that it either commission one before this threshold is finalized, or state on the record what evidence it relies upon regarding dependence liability in users with no prior opioid exposure.

6. CONSUMER WARNING INFORMATION REMOVED DURING THIS PERIOD

It has been reported, and not denied by the Department, that kratom harm information was removed from FDA web pages during 2025 following requests from a federal official, and that a further request to remove a warning regarding liver toxicity, seizures and substance use disorder was declined. That warning remains on FDA's kratom page as of 27 August 2026.

Given that the reported harm concentrates in the product category this action does not restrict, the removal of consumer information about that same category during the same period is directly relevant to this docket.

REQUESTED ACTION: Place in the docket a description of what kratom-related consumer information was altered or removed from Department web properties from 1 January 2025 to date, the requests that prompted each change, and the scientific basis relied upon.

7. PROCESS TRANSPARENCY

The notice extending this comment period states that the extension was granted "in response to a request for an extension." The requesting party is not identified.

Given the commercial consequences of where this threshold is set, and that comments submitted here are to be provided by the Secretary for consideration by the Attorney General, the identity of parties who obtained procedural accommodations should be on the record.

REQUESTED ACTION: Identify the party or parties that requested the extension of the comment period.

8. THE PRODUCTS ARE MARKETED AS NATURAL WHILE THE DEPARTMENT'S OWN ASSESSMENT

FINDS THEM SYNTHETIC

We raise one point that is independent of the threshold and within the Department's remit.

7-OH products are marketed to consumers using the vocabulary of botanicals: natural, herbal, plant based, wellness. The Department's own assessment states that the enhanced 7-OH in these products is "likely synthetically derived through oxidate chemical conversion of mitragynine isolates or kratom extracts," and that direct extraction from plant material "would simply be unfeasible economically."

Two granted United States patents describe the conversion process, US 12,492,201 B1 and US 12,466,830, "Method of converting mitragynine to 7-hydroxymitragynine." The manufacturing route is not in dispute; it is patented.

We further note the reported distribution practice of providing concentrated 7-OH as free samples to existing kratom leaf purchasers, including persons in recovery from opioid use. Where that occurs, the consumer's first exposure is made costless and the decision whether to continue is taken after dependence may already have begun.

REQUESTED ACTION: That the Department state whether marketing a chemically converted 7-OH product as a natural or plant-based botanical is consistent with federal labeling requirements, and that it refer the question to the Federal Trade Commission if it is not within FDA's own enforcement discretion.

CONCLUSION

Schedule 7-OH. The science supports it and delay serves nobody.

Then fix the citation, state the analytical method behind the threshold, disclose how the evidence base was assessed for interest, and say plainly what protection applies to the products that are staying on the shelf. A rule that restricts one company's product while leaving its competitor's unrestricted has to be able to show its work. Right now, in three specific places, it cannot.

Respectfully submitted,

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Supporting material available on request, including the full adverse event extract, Lobbying Disclosure Act filings, and the citation analysis referenced above.